

```
In [ ]: !pip install pandas matplotlib seaborn wordcloud scikit-learn

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from wordcloud import WordCloud
from sklearn.feature_extraction.text import CountVectorizer

# Pengaturan visualisasi standar agar rapi
sns.set_theme(style="whitegrid")
plt.rcParams['figure.figsize'] = (10, 6)

print("Libraries imported successfully!")
```

```
In [ ]: # Membaca dataset
df = pd.read_csv('dataset/bbc-news-data.csv', sep='\t')

# Menampilkan 5 data teratas
display(df.head())

# Menampilkan informasi dataset
print("\n--- Info Dataset ---")
display(df.info())
```

	category	filename	title	content
0	business	001.txt	Ad sales boost Time Warner profit	Quarterly profits at US media giant TimeWarne...
1	business	002.txt	Dollar gains on Greenspan speech	The dollar has hit its highest level against ...
2	business	003.txt	Yukos unit buyer faces loan claim	The owners of embattled Russian oil giant Yuk...
3	business	004.txt	High fuel prices hit BA's profits	British Airways has blamed high fuel prices f...
4	business	005.txt	Pernod takeover talk lifts Domecq	Shares in UK drinks and food firm Allied Dome...

```
--- Info Dataset ---
<class 'pandas.DataFrame'>
RangeIndex: 2225 entries, 0 to 2224
Data columns (total 4 columns):
#   Column      Non-Null Count  Dtype
---  -
0   category    2225 non-null   str
1   filename    2225 non-null   str
2   title       2225 non-null   str
3   content     2225 non-null   str
dtypes: str(4)
memory usage: 69.7 KB
None
```

```
In [3]: print("--- Pengecekan Missing Values ---")
display(df.isnull().sum())

print("\n--- Pengecekan Duplikat ---")
print(f"Total baris duplikat persis: {df.duplicated().sum()}")
```

```
--- Pengecekan Missing Values ---
category      0
filename      0
title         0
content       0
dtype: int64
--- Pengecekan Duplikat ---
Total baris duplikat persis: 0
```

```
In [4]: # Menghitung jumlah kata pada judul dan isi berita
df['title_word_count'] = df['title'].apply(lambda x: len(str(x).split()))
df['content_word_count'] = df['content'].apply(lambda x: len(str(x).split()))

# Menampilkan statistik deskriptif dari panjang teks
display(df[['title_word_count', 'content_word_count']].describe())
```

	title_word_count	content_word_count
count	2225.000000	2225.000000
mean	5.204494	378.835955
std	0.865494	238.220755
min	3.000000	84.000000
25%	5.000000	240.000000
50%	5.000000	326.000000
75%	6.000000	466.000000
max	9.000000	4428.000000

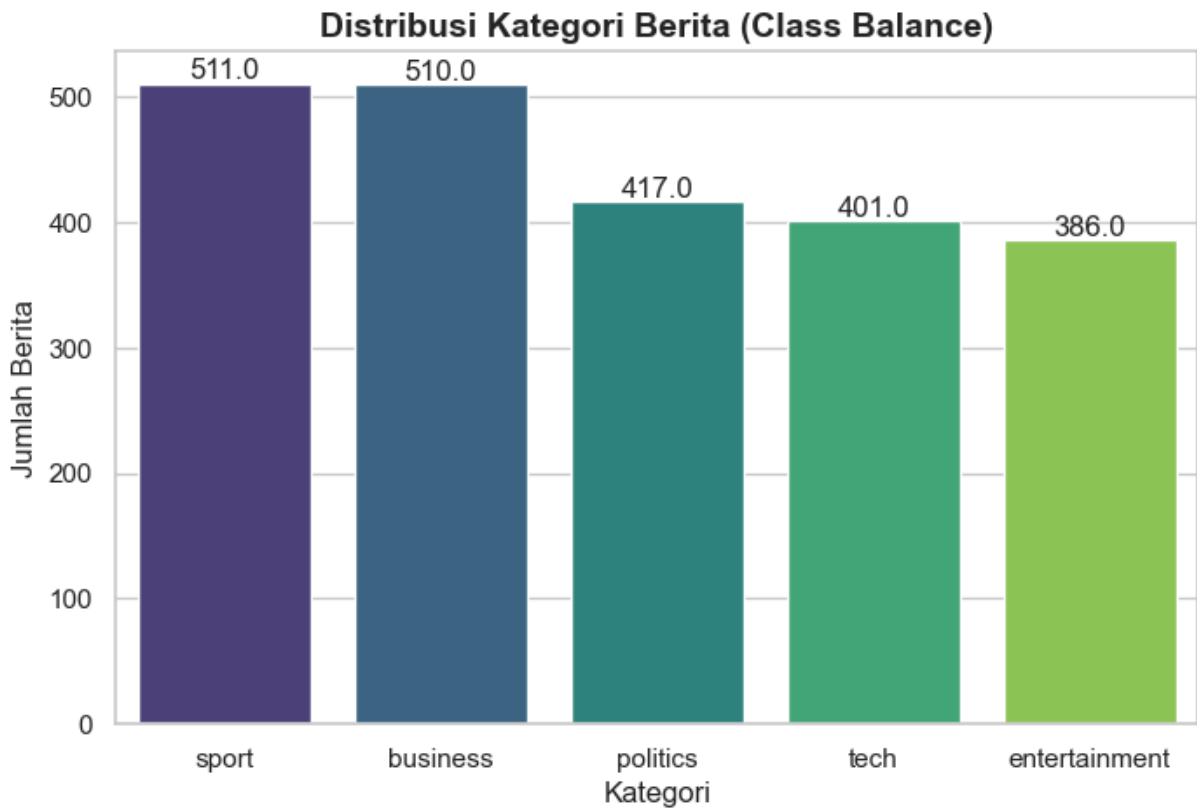
```
In [5]: plt.figure(figsize=(8, 5))
ax = sns.countplot(data=df, x='category', order=df['category'].value_counts().index)
plt.title('Distribusi Kategori Berita (Class Balance)', fontsize=14, fontweight='bold')
plt.xlabel('Kategori', fontsize=12)
plt.ylabel('Jumlah Berita', fontsize=12)

# Menambahkan angka di atas bar
for p in ax.patches:
    ax.annotate(f'{p.get_height()}', (p.get_x() + p.get_width() / 2., p.get_height()
                                     ha='center', va='center', xytext=(0, 5), textcoords='offset points')
plt.show()
```

C:\Users\axels\AppData\Local\Temp\ipykernel_9092\3664793004.py:2: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

```
ax = sns.countplot(data=df, x='category', order=df['category'].value_counts().index, palette='viridis')
```

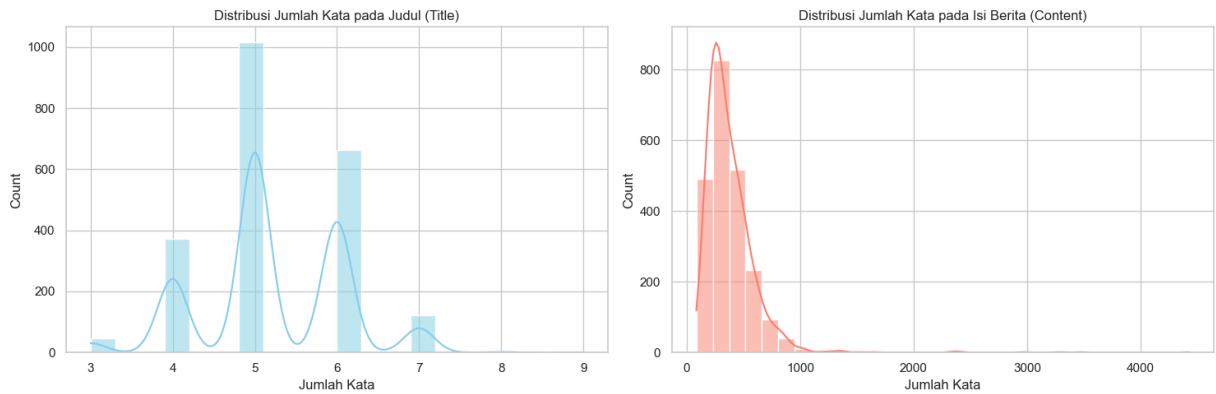


```
In [6]: fig, axes = plt.subplots(1, 2, figsize=(15, 5))

# Histogram panjang judul
sns.histplot(df['title_word_count'], bins=20, kde=True, ax=axes[0], color='skyblue')
axes[0].set_title('Distribusi Jumlah Kata pada Judul (Title)')
axes[0].set_xlabel('Jumlah Kata')

# Histogram panjang konten
sns.histplot(df['content_word_count'], bins=30, kde=True, ax=axes[1], color='salmon')
axes[1].set_title('Distribusi Jumlah Kata pada Isi Berita (Content)')
axes[1].set_xlabel('Jumlah Kata')

plt.tight_layout()
plt.show()
```

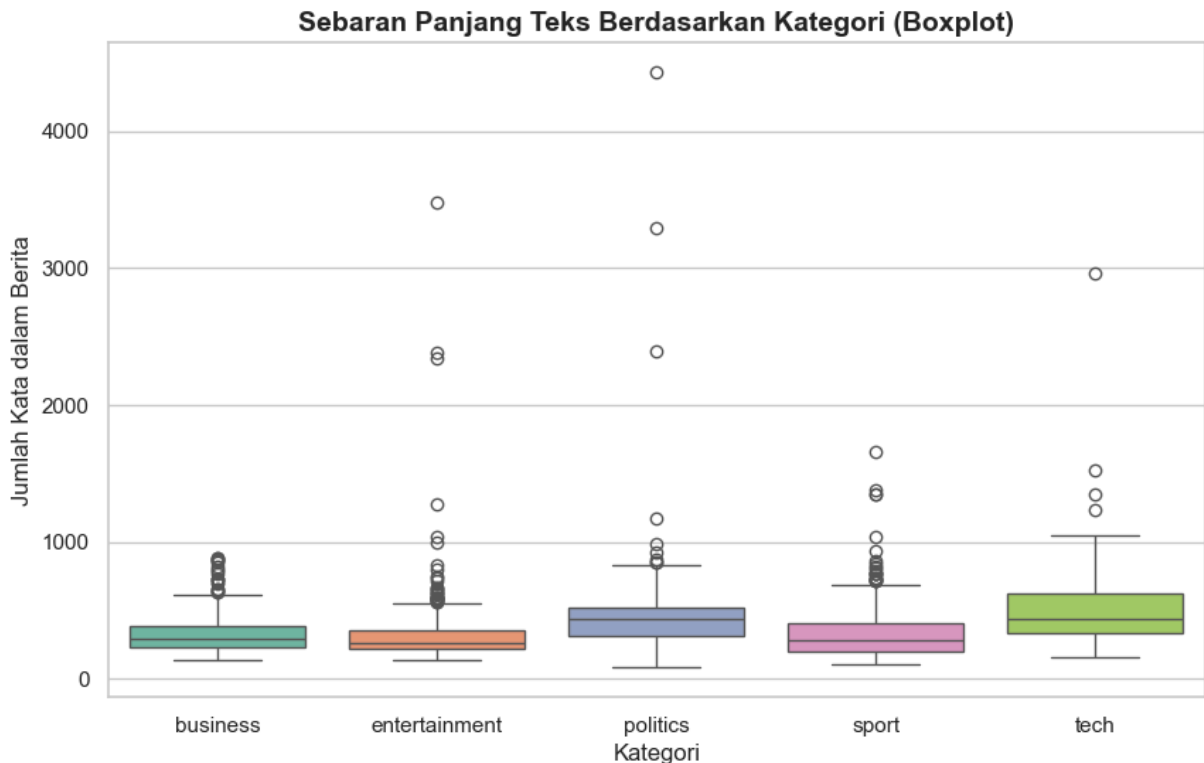


```
In [14]: plt.figure(figsize=(10, 6))
sns.boxplot(data=df, x='category', y='content_word_count', palette='Set2')
plt.title('Sebaran Panjang Teks Berdasarkan Kategori (Boxplot)', fontsize=14, fontw
plt.xlabel('Kategori', fontsize=12)
plt.ylabel('Jumlah Kata dalam Berita', fontsize=12)
plt.show()
```

C:\Users\axels\AppData\Local\Temp\ipykernel_9092\64365918.py:2: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

```
sns.boxplot(data=df, x='category', y='content_word_count', palette='Set2')
```

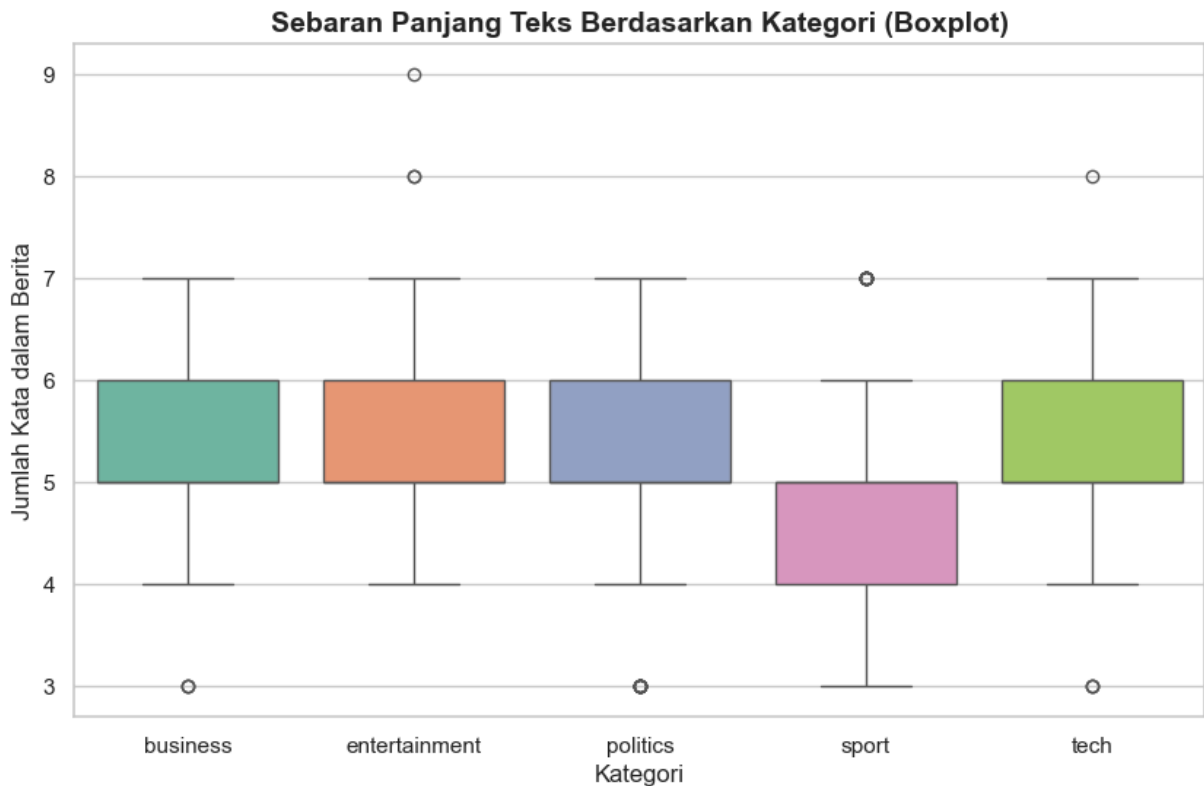


```
In [13]: plt.figure(figsize=(10, 6))
sns.boxplot(data=df, x='category', y='title_word_count', palette='Set2')
plt.title('Sebaran Panjang Teks Berdasarkan Kategori (Boxplot)', fontsize=14, fontw
plt.xlabel('Kategori', fontsize=12)
plt.ylabel('Jumlah Kata dalam Berita', fontsize=12)
plt.show()
```

C:\Users\axels\AppData\Local\Temp\ipykernel_9092\1427370683.py:2: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

```
sns.boxplot(data=df, x='category', y='title_word_count', palette='Set2')
```



```
In [8]: # Menggabungkan semua teks konten menjadi satu string raksasa
text_corpus = " ".join(df['content'].values)

# Membuat WordCloud
wordcloud = WordCloud(width=800, height=400, background_color='white', colormap='ta

plt.figure(figsize=(12, 6))
plt.imshow(wordcloud, interpolation='bilinear')
plt.axis('off')
plt.title('WordCloud: Kata-kata Terpopuler dalam Dataset BBC News', fontsize=16, fo
plt.show()
```

WordCloud: Kata-kata Terpopuler dalam Dataset BBC News



```
In [10]: def get_top_ngram(corpus, n=None, ngram=2):
# Menggunakan stop_words='english' untuk membuang kata umum (the, is, and)
vec = CountVectorizer(ngram_range=(ngram, ngram), stop_words='english').fit(corpus)
bag_of_words = vec.transform(corpus)
sum_words = bag_of_words.sum(axis=0)
words_freq = [(word, sum_words[word_idx]) for word, word_idx in vec.vocabulary_.items()]
words_freq = sorted(words_freq, key=lambda x: x[1], reverse=True)
return words_freq[:n]

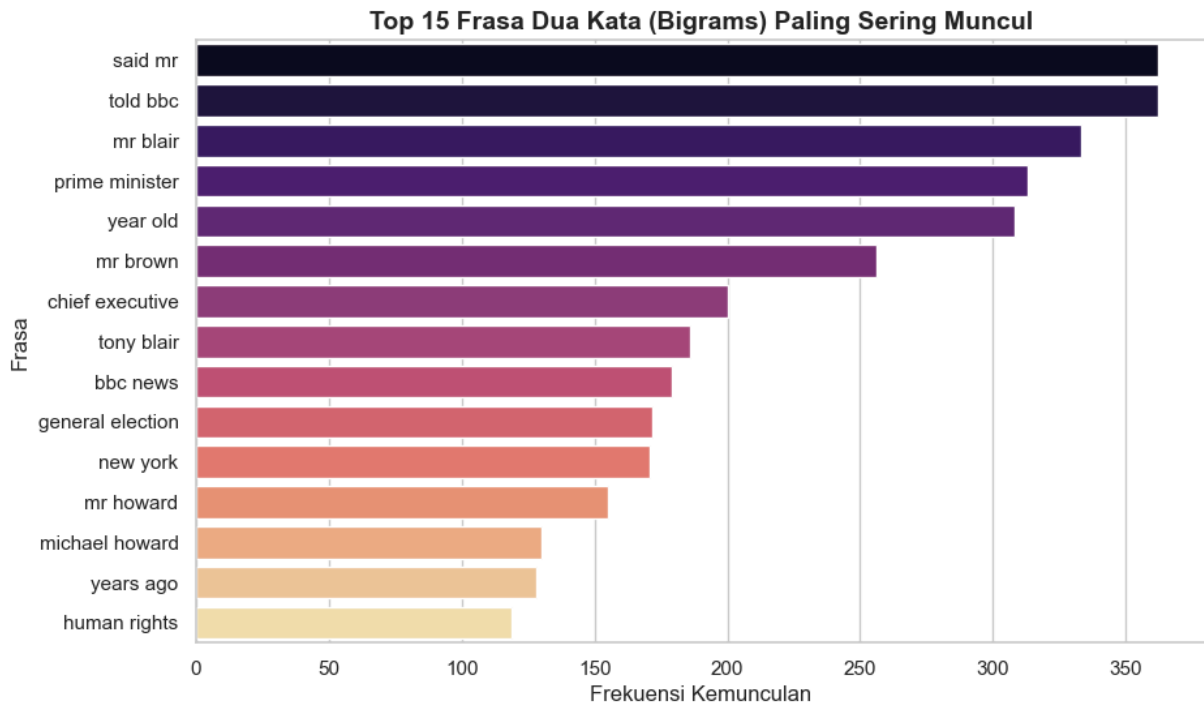
# Mengambil top 15 Bigrams (frasa 2 kata)
top_bigrams = get_top_ngram(df['content'], n=15, ngram=2)
bigram_df = pd.DataFrame(top_bigrams, columns=['Frasa (Bigram)', 'Frekuensi'])

plt.figure(figsize=(10, 6))
sns.barplot(data=bigram_df, x='Frekuensi', y='Frasa (Bigram)', palette='magma')
plt.title('Top 15 Frasa Dua Kata (Bigrams) Paling Sering Muncul', fontsize=14, fontweight='bold')
plt.xlabel('Frekuensi Kemunculan')
plt.ylabel('Frasa')
plt.show()
```

C:\Users\axels\AppData\Local\Temp\ipykernel_9092\2012977258.py:15: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `y` variable to `hue` and set `legend=False` for the same effect.

```
sns.barplot(data=bigram_df, x='Frekuensi', y='Frasa (Bigram)', palette='magma')
```



```
In [11]: # --- EDA KHUSUS UNTUK FITUR MODEL (TITLE) ---
print("\n--- Fokus pada Fitur Utama: Judul (Title) ---")

# WordCloud khusus untuk judul
title_corpus = " ".join(df['title'].values)
wordcloud_title = WordCloud(width=800, height=300, background_color='black',
                             colormap='cool', max_words=80).generate(title_corpus)

plt.figure(figsize=(10, 5))
plt.imshow(wordcloud_title, interpolation='bilinear')
plt.axis('off')
plt.title('WordCloud: Kata Kunci Terpopuler pada Judul Berita', fontsize=14, fontwe
plt.show()

# Melihat top kata pada judul
vec_title = CountVectorizer(stop_words='english').fit(df['title'])
bag_of_words_title = vec_title.transform(df['title'])
sum_words_title = bag_of_words_title.sum(axis=0)
words_freq_title = [(word, sum_words_title[0, idx]) for word, idx in vec_title.voca
words_freq_title = sorted(words_freq_title, key=lambda x: x[1], reverse=True)[:10]

print("Top 10 Kata Terbanyak di Judul:")
for word, freq in words_freq_title:
    print(f"- {word}: {freq} kali")

--- Fokus pada Fitur Utama: Judul (Title) ---
```

[illegible]

- new: 69 kali
- uk: 61 kali
- blair: 49 kali
- film: 46 kali
- tv: 41 kali
- hits: 32 kali
- set: 32 kali
- eu: 31 kali
- net: 31 kali
- deal: 30 kali